TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Detailed Study of Rotor and Other Modern Short and Long Staple Spinning Machines

A systematic machine-study sheet: how to take any spinning machine apart with the eyes — parts, specifications and material flow — worked in full on the rotor spinning machine, with parallel identification notes on the compact spinning unit and the long-staple gill box and rectilinear comb.

YM-4202 · MODERN YARN MANUFACTURING SESSIONAL · EXPERIMENT 2 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to study the rotor spinning machine in detail — identifying every part in the path of the material, stating its function, recording the machine's working specifications, and tracing the flow from sliver can to wound package — and to apply the same study method, in outline, to two further modern machines: a compact spinning unit (short staple) and the gill box and rectilinear comb of the long-staple (worsted-type) line.

1.1 The study method — the same for every machine

1. **Identify the parts in material order.** Start where the feed enters and follow the material. A part you cannot place in the flow is a part you have not understood.
2. **State one function per part,** in terms of what changes in the material across it: opened, cleaned, drafted, twisted, wound.
3. **Record the specifications that govern production:** the speeds of the working elements, the feed and delivery rates, the range of counts the machine is built for, and its drive and settings. Copy them from the machine plate and the maker's manual, and note which of them the operator can change.
4. **Trace the material flow as a diagram,** with arrows, and check the diagram against the machine: every arrow must be a real passage of material, every box a real element.
5. **Close the study with a production calculation** from the recorded specifications. A machine study that cannot predict the machine's output is incomplete.

1.2 Rotor spinning — the principle

Rotor spinning is an **open-end** system. In ring spinning the package rotates to insert twist, so the whole mass of yarn already spun is dragged round with every turn, limiting speed. Rotor spinning breaks that link. A draw-frame sliver is fed to a fast **opening roller**, whose pinned surface tears the sliver apart into individual fibres and a moving air stream. The separated fibres are carried through a transport channel into a small **rotor** spinning at very high speed. Centrifugal force throws the fibres onto the collecting surface inside the rotor, where they build into a thin ring of fibre. A seed yarn, introduced through the navel (the doffing tube), picks up the end of this fibre ring; as the yarn is drawn away, the rotation of the rotor inserts **one turn of twist for every revolution of the rotor** between the navel and the fibre ring, and new fibres arriving in the channel continuously join the open end of the forming yarn. Because twist is inserted by the small, light rotor and not by a package, delivery speeds far beyond ring spinning are possible, and the yarn is wound onto a large package directly, with no separate winding stage. The price of the principle is in the yarn: rotor yarn is a little weaker and harsher than ring yarn of the same count, with a characteristic surface of wrapper fibres, and the system suits coarse and medium counts best.

Two relations govern every rotor calculation. Twist is rotor speed divided by delivery speed:

$$\text{Delivery speed (m/min)} = \text{rotor speed (rpm)} \div \text{required twist (turns/m)}$$

and production follows from delivery speed and yarn linear density:

$$\text{Production per rotor (g/h)} = \text{delivery (m/min)} \times 60 \times \text{yarn tex} \div 1000$$

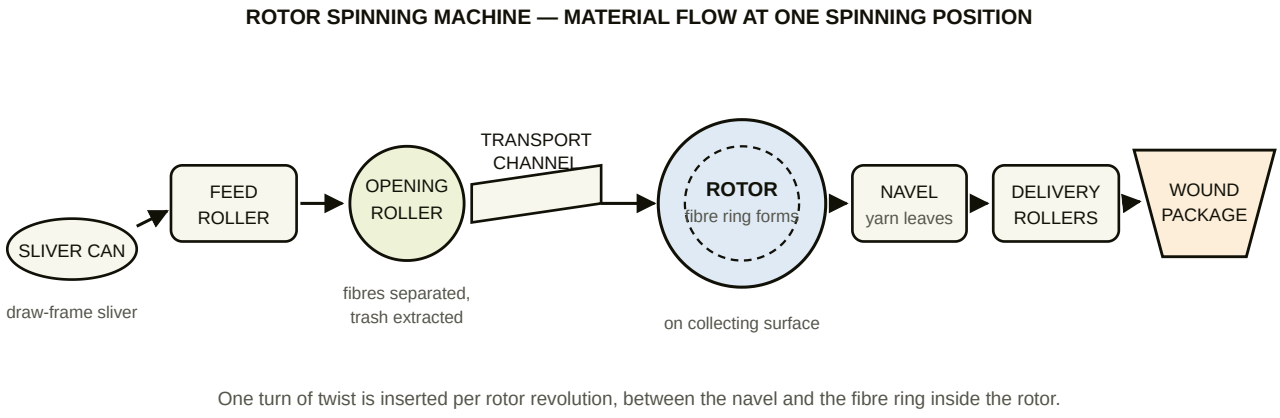


Figure 1 — Rotor spinning position, in material order. The sliver is opened to single fibres, transported by air, re-assembled as a fibre ring inside the rotor, twisted as it is withdrawn through the navel, and wound directly into a large package.

2. Apparatus and procedure

2.1 Rotor spinning machine — parts and functions

PART	FUNCTION
Sliver can and feed	Supplies draw-frame sliver to the position; the sliver is the finest feed of any spinning system, which is why rotor spinning needs no roving stage.
Feed roller and feed plate	Present the sliver, under control and at a set rate, to the opening roller; the feed rate, with the delivery rate, fixes the draft — and therefore the count — of the yarn.
Opening roller	A small, fast roller clothed with pins or saw-tooth wire. It combs the sliver fringe apart into individual fibres and, with the suction at the trash box, ejects trash and dust — the machine's cleaning point.
Transport channel	The passage through which the air stream carries the separated fibres from the opening roller to the rotor, keeping them separated and roughly aligned in flight.
Rotor	The heart of the machine: a small cup rotated at very high speed. Fibres slide down its wall onto the collecting groove, forming the fibre ring; its rotation inserts the twist. Rotor diameter and speed together set the practical count range.
Navel (doffing tube)	The guide through which the formed yarn leaves the rotor. Its surface influences yarn twist behaviour and hairiness; the yarn turns around it as it is withdrawn.
Delivery rollers	Withdraw the yarn from the rotor at the set delivery speed — the speed that, against rotor speed, fixes the twist (Section 3).
Winding unit	Builds the delivered yarn directly into a large cross-wound package, removing the separate winding operation that ring spinning requires.
Suction system	Maintains the air flow that transports the fibres and holds the spinning vacuum; a fall in suction is a fall in yarn quality before it is anything else.

2.2 Procedure for the machine study

1. **Record the machine identity:** maker, model, number of rotors (positions), and the count range stated on the machine plate.
2. **Follow one position** from sliver can to package, identifying each part of Table 2.1 on the machine itself and noting the direction of rotation and of material travel at each element.
3. **Record the working specifications** as set for the trial: rotor speed, opening roller speed, feed rate, delivery speed, rotor diameter, and the sliver hank being fed.
4. **Observe a piecing:** watch how a seed yarn is introduced through the navel and picks up the fibre ring — the clearest demonstration of the open-end principle.
5. **Collect the product:** take the package from the studied position, weigh a measured length of its yarn, and calculate the count actually spun.

6. **Complete the production calculation** of Section 3 from the recorded speeds, and compare its prediction with the machine's stated output.
7. **Repeat the identification, in outline only,** on the compact unit and on the long-staple machines of Section 2.3, noting for each the single feature that distinguishes it.

2.3 Parallel identification notes — compact, gill box, rectilinear comb

MACHINE	IDENTIFICATION POINTS AND DISTINGUISHING FEATURE
Compact spinning unit	An attachment to a ring frame's drafting zone: after the front rollers, the drafted strand passes over a perforated apron held against suction, which **condenses** it — gathering its edge fibres into the yarn body — before twist is inserted in the ordinary ring-and-traveller way. Identify it by the condensing zone and suction tube between the front roller and the yarn guide. The result is a ring-type yarn of higher strength and markedly lower hairiness.
Gill box (long staple)	The drafting machine of the worsted-type line: slivers are drafted through a bed of moving **faller bars carrying pins** (the gill), which control the long fibres positively through the drafting zone — necessary because long-staple fibres are far longer than any roller setting can grip at both ends. Identify it by the pinned faller field between feed and delivery rollers. Successive gill boxes, with doubling, prepare the sliver for combing and for the final drawing stages.
Rectilinear comb	The long-staple combing machine: a clamped tuft of fibres is combed first by a rotating circular comb and then by a straight (rectilinear) top comb as the tuft is drawn forward, removing short fibre (noil) and leaving the long fibres parallel. Identify it by the nipper/clamp assembly, the circular comb cylinder and the detaching rollers. It does for long staple what the cotton comber does for fine cotton, on a tuft-by-tuft intermittent cycle.

SPECIFICATION (RECORD FOR THE STUDIED MACHINE)	SAMPLE ENTRY	OWN READING
Rotor speed	90,000 rpm	—
Required twist	750 turns/m	—
Yarn spun	25 tex	—
Delivery speed (calculated)	120 m/min	—

The sample entries above are the values worked in Section 3.

3. Worked calculation

WORKED CALCULATION — ROTOR DELIVERY SPEED AND PRODUCTION

Rotor speed = **90,000 rpm**; required twist = **750 turns/m**; yarn = **25 tex**.

Each rotor revolution inserts one turn, so the yarn must be drawn away at:

Delivery = rotor speed ÷ twist = 90,000 ÷ 750 = **120 m/min**.

Length delivered per rotor per hour = 120 × 60 = 7,200 m.

Production per rotor per hour = 7,200 × 25 ÷ 1000 = **180 g per rotor per hour**.

Extended: per rotor per 8-hour shift = 180 × 8 = 1,440 g = **1.44 kg**; a 100-rotor machine at full efficiency would therefore produce 1.44 × 100 = **144 kg per shift**, and at 95% efficiency 144 × 0.95 = **136.8 kg per shift**.

For reference, the same yarn is 590.5 ÷ 25 = **23.6 Ne**, and its twist of 750 turns/m is 750 ÷ 39.37 = **19.1 turns per inch**.

The advantage is arithmetic, not opinion: a delivery of 120 m/min is several times that of a ring frame on a comparable count, whose delivery is capped by traveller speed and package size.

4. Precautions

- Study a running machine from the operator's aisle only; never open a rotor housing or reach into a spinning position while it is running — the rotor's speed makes it effectively invisible and unforgiving.
- Copy specifications from the machine plate and manual, and mark clearly which figures are plate values and which are the trial's settings; mixing the two corrupts every later calculation.
- Check suction before judging yarn quality at a demonstration position: a partially blocked suction line gives a weak, uneven yarn that says nothing about the machine's design.
- When tracing material flow, confirm each arrow against the machine. The transport channel in particular is easily drawn in the wrong sense; the fibres travel with the air stream, from opening roller to rotor.
- Handle a removed package by its ends, not by its wound surface, and weigh test lengths promptly — the count check of step 5 in the procedure depends on an undamaged sample.
- On the long-staple machines, treat the pinned fallers and combs as the hazard they are: pins are identified with the machine stopped, never by touch on a moving gill bed.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is rotor spinning called an open-end system, and what limitation of ring spinning does it remove?

2. A rotor runs at 100,000 rpm and the required twist is 800 turns/m. Calculate the delivery speed, and the production per rotor per hour for a 20 tex yarn.
3. State the function of the opening roller, and name the second job it performs besides opening.
4. Why does rotor spinning need no roving stage, and which machine upstream therefore feeds it directly?
5. What is the distinguishing feature of a compact spinning unit, and what yarn property does it chiefly improve?
6. Why does the long-staple line need a gill box, when the cotton line drafts satisfactorily with rollers alone?

Answer key

1. Because the feed material is completely broken open — the sliver is separated into individual fibres — and the yarn is re-formed from those fibres at an open, continuously renewed end inside the rotor, rather than being twisted as one continuous strand from feed to package. It removes the ring frame's speed limit: in ring spinning the package itself must rotate to insert twist, and traveller speed and package mass cap the delivery; in rotor spinning only the small, light rotor turns.
2. Delivery = 100,000 ÷ 800 = **125 m/min**. Production = 125 × 60 × 20 ÷ 1000 = **150 g per rotor per hour**.
3. The opening roller tears the fed sliver apart into individual fibres, which the air stream can then transport to the rotor. Its second job is cleaning: trash and dust liberated at its surface are extracted at the trash box, making it the machine's cleaning point.
4. Because its feed is sliver, not roving: the opening roller can accept a draw-frame sliver directly and perform, by opening, the first stage of separation that other systems approach through successive drafting stages. The draw frame therefore feeds the rotor machine directly, and the simplex and its roving bobbins disappear from the line.
5. Its distinguishing feature is a condensing zone — a perforated apron under suction — placed after the front drafting rollers, which narrows the drafted strand and draws its edge fibres into the yarn body before twisting. It chiefly improves hairiness (markedly lower) and, with it, yarn strength and surface appearance.
6. Because long-staple fibres are longer than the distance over which a roller drafting zone can control them: a fibre gripped at neither end, or released early, moves uncontrolled and creates thick and thin places. The gill box's moving bed of pinned fallers carries and controls the fibres positively through the whole drafting field, which rollers alone cannot do at that fibre length.
